(a) Methodological Ground Simulation Standards (3-Chamber Parity)

Chamber 1: Geomagnetic Baseline (GMF)

Magnetic Regime: 45–50 μT Ambient Terrestrial Field

Experimental Purpose: Terrestrial environmental control (calibrated local vector)

Chamber 2: Near-Null Field Simulation (NNMF)

Magnetic Regime: <10 nT (Shielded / Actively Cancelled)

Experimental Purpose: Lunar surface magnetic simulation (<0.8 nT ambient void)

Chamber 3: Matched Bifilar Sham Control

Magnetic Regime: Opposing windings; net B = 0

Experimental Purpose: Controls for ohmic heat, acoustic vibration & power-supply ripple

(b) In Situ Causal Reversal Protocol (Proof of Concept)

Three-Step Causal Proof Inside Magnetic Shielding

Step 1: Hypomagnetic Exposure

<10 nT inside μ-metal shield

Elicits emergence lag (Δt50 = 12–36 h), chlorosis & ROS leakage

Step 2: Vector Re-application

45 μT active vector restored in situ

Re-applied via internal triaxial Helmholtz coils without disturbing microclimate

Step 3: Biological Phenotype Rescue

Phenotype restored to GMF baseline

Wild-type emergence & greening speed recovered; proves magnetic causality

Definitively rules out microclimatic, ethylene, or enclosure artifacts

(c) Post-Flight Phenotypic Risk Disambiguation Framework

Operational Multi-Stressor Disambiguation Matrix for Space Plant Biology

Near-Null Magnetic Field (<0.8 nT)

Primary Biophysical Target:

Cryptochrome flavin radical pairs (FAD/TrpH•+);
mitochondrial [4Fe-4S] cluster electron transfer.

Diagnostic Phenotypic Markers:

- Emergence lag (Δt50 = +12 to 36 h); capacity -10 to -30%
- Repressed chlorophyll (HEMA1, POR) & anthocyanin (CHS)
- Mitochondrial ROS leak; induction of AOX1D & CAT3
- Normal nuclear DNA integrity (zero γ-H2AX foci)

Disambiguation & Rescue Strategy:

- Correlate with sample-level 3-axis canopy magnetometry
- Fully rescued by 45 μT vector re-application
- Completely insensitive to 1g centrifuge spinning

[Diagnostic: Rescued by 45 μT | Insensitive to 1g]

Lunar Partial Gravity (0.166 g)

Primary Biophysical Target:

Columella statolith sedimentation kinetics;
mechanosensitive ion channels (MSL); PIN polar auxin.

Diagnostic Phenotypic Markers:

- Germination capacity & greening rate are normal
- Directional root skewing, waving, & loss of g-angle
- Cell wall remodeling (pectin demethylesterification)
- Normal CRY signaling; localized apoplastic ROS (RBOH)

Disambiguation & Rescue Strategy:

- Fully rescued to Earth phenotype by 1g centrifuge
- Differential 1g vs μg controls in flight centrifuge
- Completely unaffected by magnetic vector modulation

[Diagnostic: Rescued by 1g | Insensitive to B-field]

Deep-Space Radiation (GCR / SPE)

Primary Biophysical Target:

Nuclear & organellar DNA double-strand breaks;
radiolytic hydroxyl free radical (OH•) overflow.

Diagnostic Phenotypic Markers:

- Severe DNA double-strand breaks & γ-H2AX foci
- ATM/ATR activation; chromosome aberrations & G2/M arrest
- Widespread systemic oxidative stress & lipid peroxidation
- Emergence fails only at acute lethal doses (>50 Gy)

Disambiguation & Rescue Strategy:

- Independent of magnetic field or gravity vector
- Strictly correlates with canopy radiation dosimeters
- Mitigated by shielding & chemical radioprotectants

[Diagnostic: Correlates with Dosimeter | DNA Breaks]